

Section 1 - BRIDGE SUPERSTRUCTURE

PC/PS PARTIAL-DEPTH CONCRETE DECK PANEL (MISCELLANEOUS DETAILS)

XS Sheet Numbers:

xs1-180-3 (xs1-180-3 shall be used in conjunction with xs1-180-1 or xs1-180-2)

Description of Component:

PC/PS Partial-Depth Concrete Deck Panel (Miscellaneous Details)

Standard Drawing Features:

Plan – Rectangular End Panel

1. Rectangular end panel is similar to the rectangular panel.
2. Panel rebar extension is shown at locations of the abutment or the bent.

Plan – Skewed End Panel

1. Panel length "L" is the same as "L" from xs1-180-1 or xs1-180-2.
2. Panel width "W" is limited to 4'-0" min. and 12'-0" max. and needs to be shown on the contractor's shop drawings.
3. Maximum skew of the panel is 20°.
4. Strand spacing "S" is the same as "S" from xs1-180-1 or xs1-180-2.
5. The strand spacing "S" at the skewed side should take into account the 2 ½" (min.) to 4" (max.) distance from the edge of the panel to the centerline of strands.
6. Strands with a bonded embedment length shorter than 3'-6" shall be replaced by rebars, as shown.

Section B-B

This section shows Detail 1 and Detail 2 locations.

Section C-C

This section shows the reinforcement details at the skewed corner.

Detail 1

1. Strands shall be located at the mid-depth of the panels (i.e., at the center of the gravity of the panels).
2. The side of the skewed edge of the panel shall be perpendicular to the bottom soffit.

Detail 2

DETAIL 2 shows the dimensions of the key and reinforcement details. The key's dimensions may differ slightly for various deck panel depths and per precast manufacturers.

Section at Deck Panel Joint

The reinforcement for the CIP deck is not permitted to sit directly on PC/PS deck panels. There is a minimum of 1" required clearance between deck reinforcement and the top of PC/PS deck panels.

Design/General Notes

The deck panel design and details are based on AASHTO LRFD Bridge Design Specifications, 8th Edition, with CA Amendments. Section 51-4.02D(7) and Section 51-4.03G "Deck Panels" of Caltrans Standard Specifications provide information on PC/PS deck panel fabrication and construction. Designers must read these sections of the Standard Specifications.

Additional Drawings Needed to Complete PS&E:

This sheet works together with XS Sheet No. xs1-180-1 or xs1-180-2.

Contract Specifications:

Standard Specifications 2023

Restrictions on Use of Standard Drawings:

The PE is responsible for applying the pre-designed Table 1 from xs1-180-1 or xs1-180-2 User Guide to the bridge.

Special Considerations:

Under some circumstances, PDPs are not permitted to be used. See STP 9.1. The PE must determine if using xs1-180-1, xs1-180-2, and xs1-180-3 sheets is appropriate for the bridge. The PE shall stamp the xs1-180-1, xs1-180-2, and xs1-180-3 sheets with a valid California Professional Engineer License Stamp.